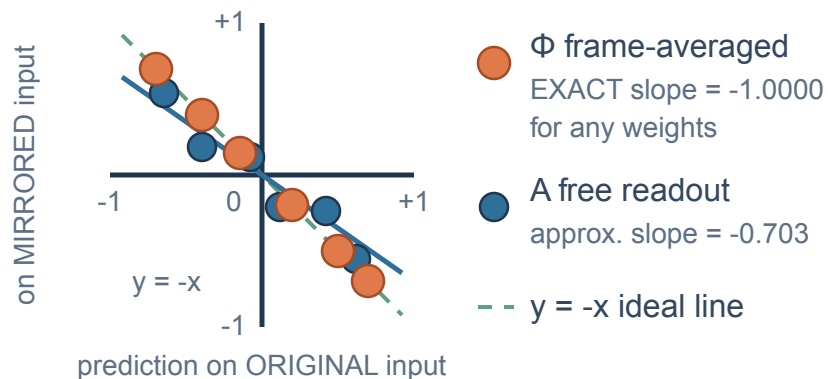


Built-in geometry beats hoping it emerges

Imposing the antisymmetric constraint by construction vs hoping a free readout learns it. Measured; transductive.

(a) Exactness of the mirror

Decoded scalar: mirrored vs original prediction, against $y = -x$.

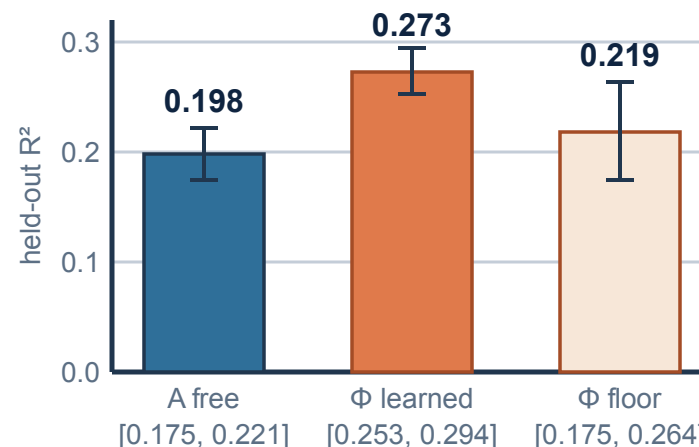


Frame-averaged ENCODER (E3a):

token-level reflection-equivariance error = 0.0
(exact, zero retraining).

(b) Decodability (held-out R^2 , 95% CI)

626 modeled; repeated source-disjoint CV (10 seeds).



Φ (learned) vs A: non-overlapping CIs $\rightarrow$ the antisymmetric constraint **UNLOCKS** decodability the free readout misses.

Control: Ψ (symmetric half)

$R^2 = 0.015$, slope +1.0.

Cannot predict the antisymmetric target.

Control: free MLP ($\geq$ capacity)

$R^2 = 0.047$, slope -0.43.

No shared-weight tie $\rightarrow$ the gain is the CONSTRAINT, not capacity.

Honest boundary

Still far below raw ceiling ≈ 1.0 ;
sign consistency 0.57 (< 0.75) $\rightarrow$
recovered axis is partial.

All results are transductive (internal validity only; the encoder was trained on the evaluated sequences).

Source video is the independent unit; folder labels are dataset annotations, not diagnoses.